Requirements Gathering

Requirements Gathering

- **Requirements gathering** is the practice of researching and discovering the requirements of a system from users, customers, and other stakeholders
  - Also known as ***requirements elicitation***

- The term 'requirements gathering' is more commonly used in industry, but 'requirements elicitation' is often used in textbooks and academia... why?
  - Requirements gathering implies we are simply collecting information; in practice we often need to much more pro-actively discover requirements

Definition Source: https://en.wikipedia.org/wiki/Requirements_elicitation

Requirements Gathering Practices

- Requirements gathering may involve many data collection practices, such as but not limited to...
  - Background research
  - Interviews
  - Surveys
  - User Observations (e.g. ethnographic studies)
  - Brainstorming

- Exact practices used depend on the system being developed and the related process

Requirements Gathering Practices

- Requirements gathering involves analysis, the understanding and researching of topics like...
  - The problem domain
  - The stakeholders and their relationships
  - Users
  - Tasks users need to be able to perform
- Some of these ideas overlap with design of the system
- Sometimes we can use formal processes and practices, but sometimes it's primarily about asking and finding answers to the right questions

Analysis vs. Data Collection

- Analysis is about the things we want to know...
  - Problem domain
  - Stakeholders
  - Users
  - Tasks

- Data collection practices are how we come to know these things...
  - Background research
  - Interviews
  - Surveys
  - User observations
  - Prototyping

Requirements Gathering

- Requirements gathering is cyclical...
  - Collecting data with which to perform analysis using formal practices (interviews, surveys, etc.)
  - Performing analysis to better understand aspects of the system (domain, users, tasks, etc.)
  - Creating specifications which formalize and represent what will be built (requirements, mockups, etc.)
- We might then ***use*** those specifications to help collect more data, with which we perform more analysis, to come up with better specifications

Let's look at analysis first...

i.e. all the things we want to know more about...

Domain Analysis

- The **problem domain** is the specific area (or domain) of the user's need (or problem) that is within the scope of the new system

- A subject or topic that someone could be an expert in
  - Medicine
  - Manufacturing
  - Credit card transactions

- Effectively the problem domain is the set of all the things and ideas that will be involved in the system

Domain Analysis

- The domain can be narrowed down from extremely broad to very specific

- The domain focus can be narrowed like this...
  - The web
  - E-commerce
  - Online restaurant ordering
  - Use of online shopping carts
  - Integration to local kitchen systems

Domain Analysis

- Who understands the domain(s)?
  - Client
  - Contractors
  - Employees
  - External support

Domain Analysis Objectives

- Understand all terms and concepts relevant in the problem domain for the system
- Understand the system as-is (replacement)
- Understand the space for the system (new)
- Identify problems and opportunities that call for a new system
- Understand the organization(s) the system is created for
- Discover varying points of view

Studying The Current State

- Who is the organization(s) the system is for?
  - What are its objectives?
  - What are its responsibilities?
- Who are the stakeholders?
  - Who can describe the system?
  - Group by role: external, internal, operational and executive
- How does current system run, if it exists?
  - Replacement or new?
  - Current state

Studying The Organization

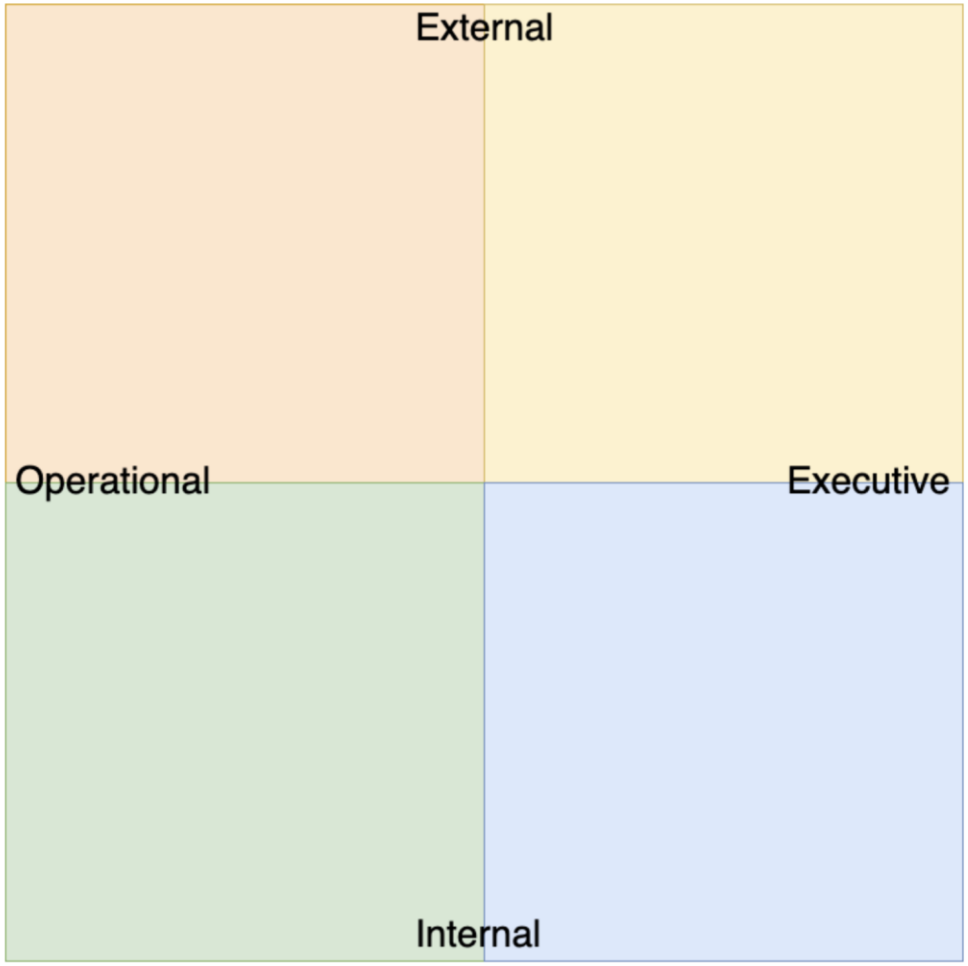
- What is the size of the organization? (i.e. large, medium, small)
  - Large usually implies more established, more systems to integrate with
  - Small usually means more flexible, better able to adapt to new ways of thinking/working
- Is the organization international? (think: laws)
- What is the attitude of the organization?
  - Do they take pride in taking risks or in giving customers the most stable experience? Can dictate reliability requirements
- What is the style of the organization's current systems?
  - Often dictates both usability and performance requirements

Who Are The Stakeholders?

- Who wants it to work one way?
  - Who is using it?
  - Who is profiting from it?
  - Whose work is affected by it?
- Usually many
- Try to pick minimum set over all groups
  - Never just the managers!
  - Why not?

Stakeholder Categorization

- How to decide which stakeholders to include?
- Attempt to categorize

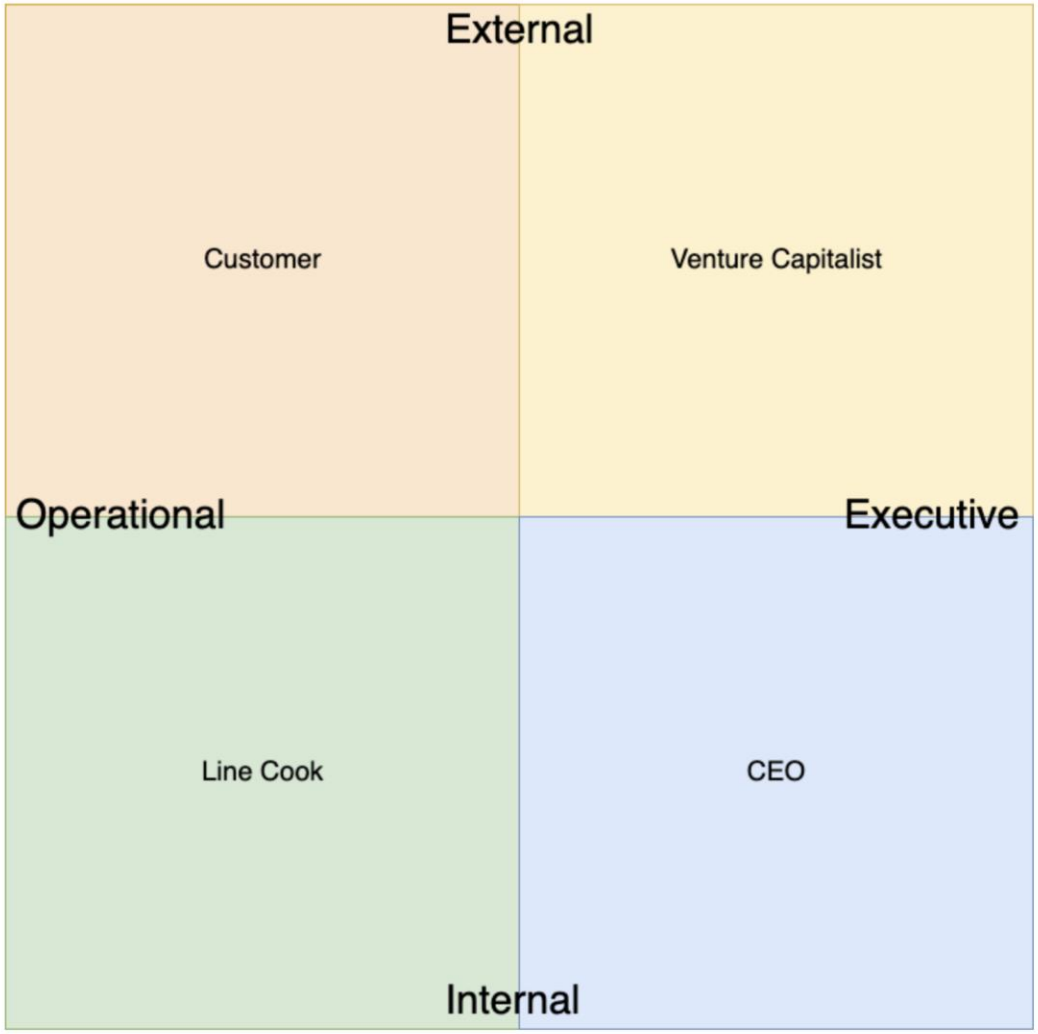


Stakeholder Categorization

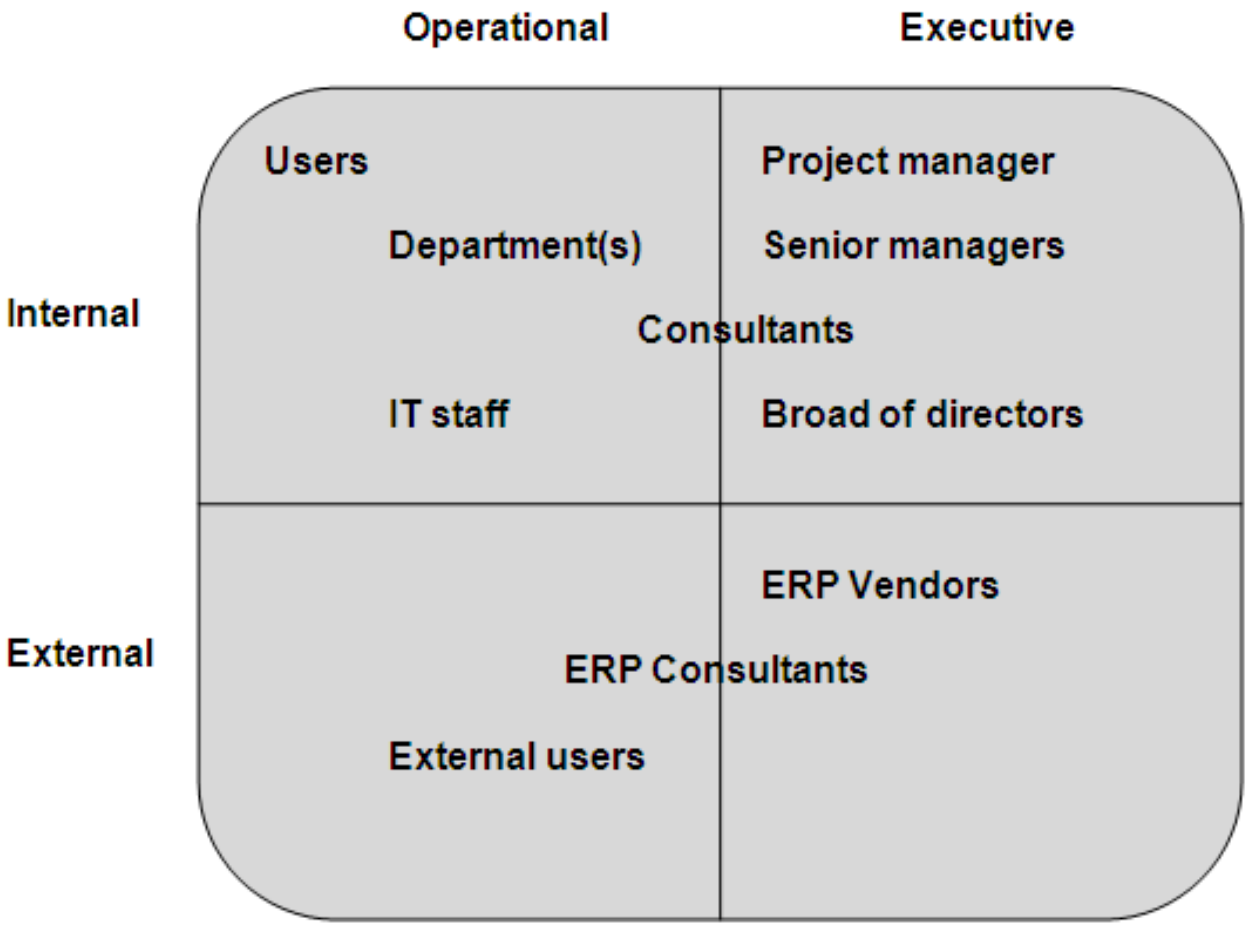
- Internal
  - Inside the organization
- External
  - Outside the organization
- Operational
  - Directly interacts with the system
- Executive
  - Indirectly interacts with the system

Stakeholder Categorization

- Usually stakeholders will be in multiple categories
- Could anyone fit the role of being both executive and operational?



Another Example…



Source: https://www.researchgate.net/publication/281781473/figure/fig2/AS:614275102044163@1523466005883/Types-of-stakeholders.png

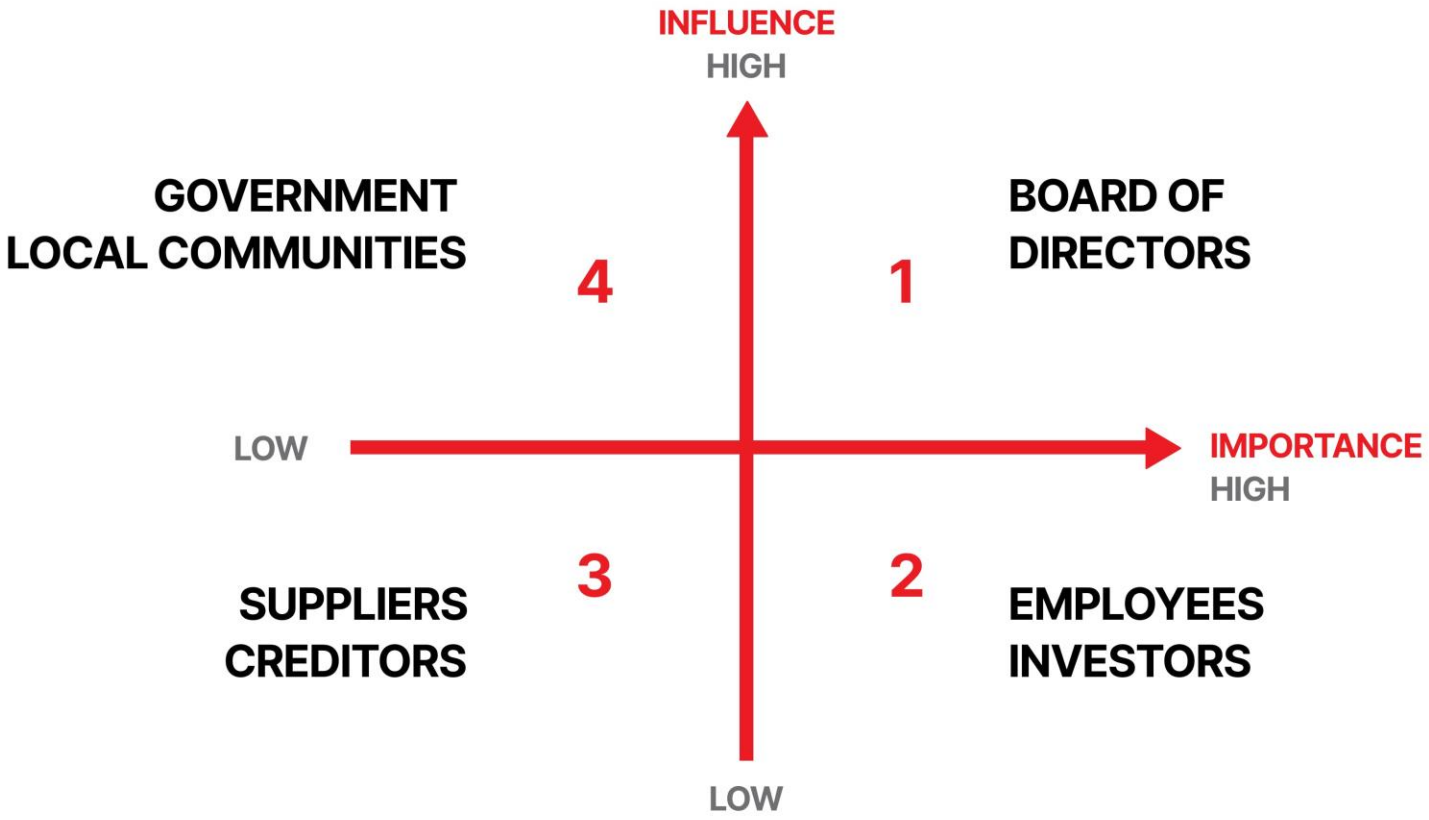
Stakeholder Categorization

- Categories help us to produce and brainstorm who are the stakeholders
- Who will use the system to find operational stakeholders
- Who will experience benefits or drawbacks of the system or otherwise be responsible for the system to find executive stakeholders
- Then easy to categorize as internal/external

Stakeholder Analysis

- Now, we have a list of stakeholders, but need to determine their interest...
  - Decide who they are
  - How is their position influencing what they want?
  - What might their biases be?
  - What kind of influence do they have on the system? In other words, how important are they to the system's ultimate acceptance and longevitiy?

We can create similar grids for other attributes like influence & importance…



Source: https://images.prismic.io/superpupertest/26c5387b-81b8-4cd0-8332-45ad790f2811_Internal6.png?auto=compress,format

Stakeholder Analysis

- Can go further than basic classification after identifying many stakeholders...
  - What do they know
  - Domain expertise
  - Exposure to issues
  - Importance in system acceptance
  - Personal objectives

Stakeholder Analysis Example

Line Cook

- High knowledge of day to day activities, how kitchen system is currently used
- Domain expertise (restaurant)
- Significant exposure to system issues
- Little influence for system acceptance
- Objective is to keep customers happy by producing enough food to fulfill all orders

Restaurant Owner

- High knowledge of business; low knowledge of day to day activities
- Domain expertise (business)
- Low exposure to system issues
- High level of influence for system acceptance
- Objective is to maximize profit

Stakeholder Analysis Example

- So which of these two stakeholders should we focus on interviewing?

- **Both!**
  - May want to say owner, since they have high importance in accepting the system
  - But if system does not work for employees, they won't use it, meaning no benefit to the owner

Current System State

- Need to know about current (and desired) system
- How to get answers?
- Stakeholders!
  - Start big, work small
  - Discover what to ask
  - Know what you don’t know

Current System State

- Basic questions to get started

- Is there a system right now that does this job?
  - What are its tasks?
  - Does it have side effects?
  - What's wrong with it?
  - Which stakeholders are most affected by change?

- Is the system a brand new addition?
  - What is missing?
  - What is the impact?
  - Is a non-software system being used and what does it look like? Otherwise, how do users currently get around the lack of system?

User Analysis

- Users are stakeholders, but not all stakeholders are users. Users are the operational stakeholders.
  - We need to analyze users specifically because they will impact the system requirements and design.

- **User analysis** is about determining what characteristics of our users may influence the requirements and design of a system
  - User analysis tends to drive interface and usability requirements and design in particular

- What do we mean by characteristics of our users?
  - Technical abilities, personalities, gender/age/etc.

User Analysis

- It is closely related to, but distinct from **task analysis,** where we analyze how user tasks are performed to influence the design of our system

- User and task analysis are often done together to inform and guide user interface design

- User and task analysis occur before design takes place

- Users and their tasks can vary in many different ways!

In what ways can users be different from one another?

User Diversity

- Abilities – physical, cognitive, motor skills, etc.
- Backgrounds, prior knowledge
- Socioeconomics (e.g. income level)
- Motivations
- Personalities
- Cultures
- Disabilities
- Work styles
- Age, gender, height, weight and other characteristics
- Device type (desktop, tablet, mobile, etc.)

- All of these characteristics and more can influence a system's requirements and desgin

We can use user characteristics to influence requirements…

Physical Abilities

- Anthropometry studies human body measurements
  - e.g. 5th to 95th percentile for height, weight, etc.
  - How would someone with a 99.99th percentile hand size do with the iPhone 5?

- Perception abilities
  - Vision-related: low vs bright light, colours, response times, depth perception, visual aids (or not), etc.
  - Similar concerns related to hearing abilities



iPhone 5 advertisement explaining iPhone device size was built so the touchscreen could be accessed with one finger: https://www.youtube.com/watch?v=9W7qquvhjNM

Cognitive Abilities

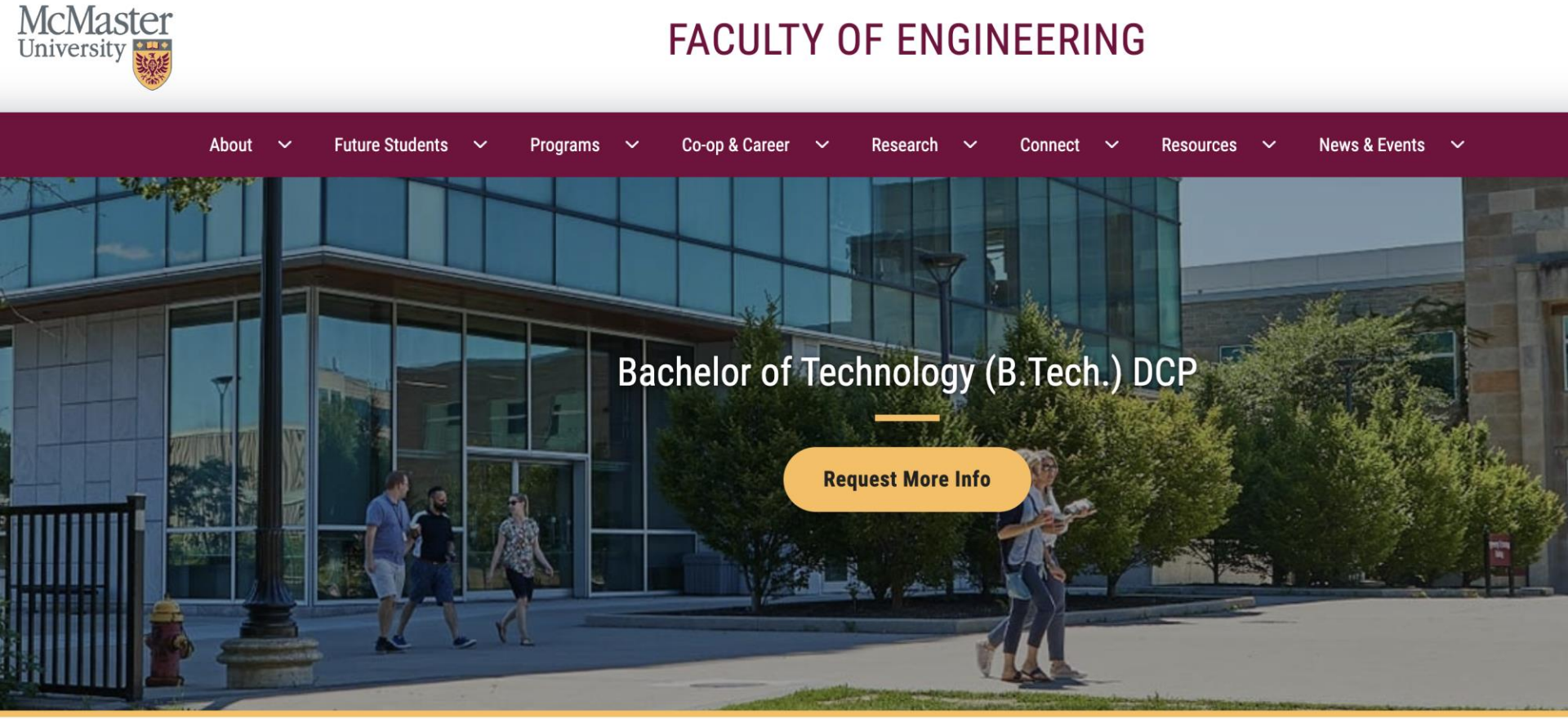
- Classification by the journal of Ergonomics Abstracts
  - Short-term and working memory
  - Long-term and semantic memory
  - Problem solving and reasoning
  - Decision making and risk assessment
  - Language communication and comprehension
  - Search, imagery, and sensory memory
  - Learning, skill development, knowledge acquisition, and concept attainment

- Above concepts used more than general idea of 'intelligence', which has a more contentious definition and ability to measure

Cognitive Abilities

- Journal of Ergonomics Abstracts also recognizes cognitive abilities can be affected by...
  - Arousal and vigilance
  - Fatigue and sleep deprivation
  - Perceptual (mental) load
  - Knowledge of results and feedback
  - Monotony and boredom
  - Sensory deprivation
  - Nutrition and diet
  - Fear, anxiety, mood, and emotion
  - Drugs, smoking, and alcohol
  - Physiological rhythms

Why are good website menus generally limited to 7 or so options?



Cognitive Abilities

- Most adults can hold 7 ± 2 items in their short term memory (Miller, 1956)
  - Makes a range of 5-9 menu items ideal
  - Anymore than that and people will forget the first items they read by the time they read the last one
  - Use submenus, but also limit those to 5-9 items

- This applies to more than website menus...
  - Ever been to a restaurant with so many options, even on subcategories, that you can't decide?
  - Some smart restaurants apply the same principles to their menu design

Personalities

- Like intelligence, different taxonomies and ways to measure, lively debates
  - e.g. Myers–Briggs, allegedly discredited but widely used...
    - Introversion/Extroversion
    - Thinking/Feeling
    - Sensing/Intuition
    - Judging/Perceiving
  - Online tests to classify you as INTJ, ENTP, etc.

- Some systems allow user multiple ways to organize information
  - e.g. Inbox organization... a big messy inbox or neatly organized into categories? Up to user.



CATS: ALL YOUR BASE ARE BELONG
TO US.

Cultural

- Beyond language translation, there are many other concerns when presenting information:
  - Titles like Mr., Ms., etc.
  - Left-to-right vs right-to-left reading
  - Measurements (weight)
  - Dates
  - Etiquette, tone, formality
  - Metaphors

Users With Disabilities

- Vision impairments
  - Screen readers
  - Refreshable braille displays
  - Magnification
- Hearing impairments
  - Captioning on videos
- Motor impairments
  - Speech recognition
  - Eye-tracking
  - Special input devices instead of keyboards/mice/etc

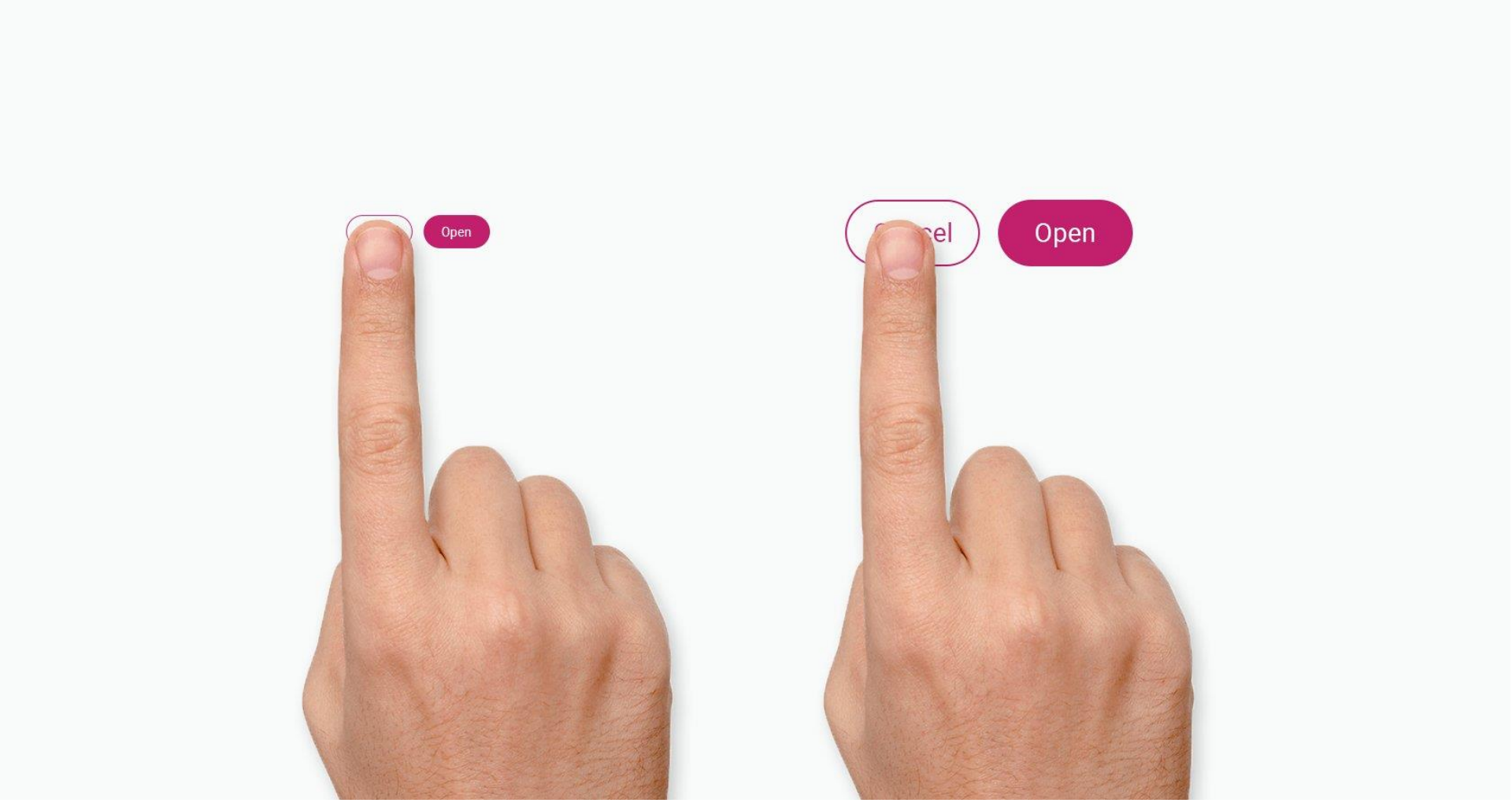
Accessibility Guidelines

- More than using the "alt property" with images!

- Web Content Accessibility Guidelines (WCAG)
  - https://www.w3.org/WAI/standards-guidelines/wcag/
  - "Except for captions and images of text, text can be resized without assistive technology up to 200 percent without loss of content or functionality."

- Accessibility for Ontarians with Disabilities Act (AODA)
  - https://www.essentialaccessibility.com/blog/aoda-website-compliance
  - Ontario organization websites must comply with WCAG by certain timelines

Who might larger buttons be useful for?



Source: https://bs-uploads.toptal.io/blackfish-uploads/uploaded_file/file/22099/image-1560959747336-8eae72a913f6f0ccf6d03a0decf4938d.png

Age

- Seniors
  - Physical, cognitive abilities generally declining
- Accommodating seniors
  - Larger font-sizes
  - Larger buttons
  - Text description underneath icons
    - Icons meaningful to 20-40 year old people may not be meaningful to 90-year-olds
  - Touchscreens over mouse interface
  - Autocomplete word/search suggestions
    - This was originally inspired by accommodating seniors
  - Lower cognitive load
    - Present less options, less information, simplified layouts

Age

- Children
  - Age-limited abilities, e.g. reading, comprehension
- Accommodating children
  - Fostering curiosity, intrinsic motivation
    - Explorable interfaces that give children control
    - Try again without penalties
  - Safety concerns
    - Does the system allow for communication?
  - Dexterity and other abilities
    - Can they be expected to drag, double-click, etc?
    - How long can they be expected to pay attention?

Gender

- "no clear pattern of gender-related preferences in interfaces has been documented"
  - Shneiderman, Designing the user interface (2016)

- Certain video games will draw higher number of female users than male users and vice versa
  - e.g. The Sims, Candy Crush Saga, Farmville draw more female players

Devices

- High speed vs low speed Internet access
  - Are users with lower speeds accommodated?
  - Techniques exist to allow web server to send smaller images to certain users
- Display size and resolution?
  - Smartphones with 3.5" screens vs desktops with 27" screens
  - Resolutions from 640x480 to 4K
  - Responsive design can help here...
- Inputs: touchscreen vs. mouse and keyboard
  - Recommended minimum sizes for "tappable" buttons

Task Analysis

- **Task analysis** is the analysis of how user tasks are performed in order to influence system requirements and design

- User and task analysis are often done together to inform and guide requirements and design
  - Users and their tasks can vary in many different ways!

- Though technically distinct concepts, parts of user analysis very much inform task analysis and overlap with task analysis

The following user characteristics are especially connected to task analysis...

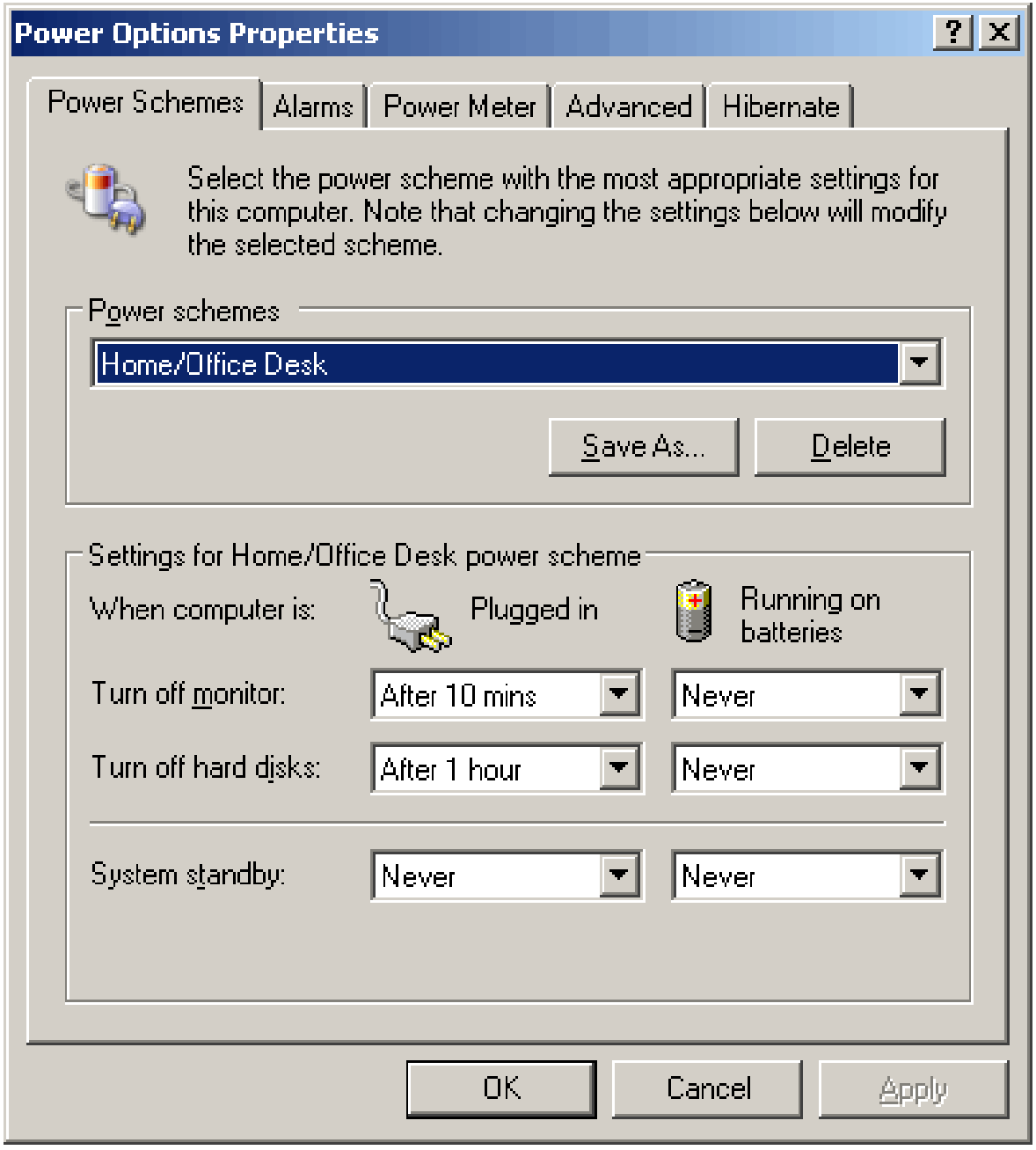
User Domain Knowledge

- How much expertise does the user have that is relevant to the interface's problem domain?

- e.g. video editing problem domain
  - iMovie comes with Mac OS X
    - Allows for simple editing, can cut clips and place clips one after the other to create larger videos
    - For users making family movies, maybe YouTube videos
  - Final Cut Pro
    - Syncing up 64 camera angles of the same scene to pick the best one... did you even know that was a thing?
    - You're probably not a professional video editor. It was used to make Parasite, Best Picture Academy Award winner in 2019!

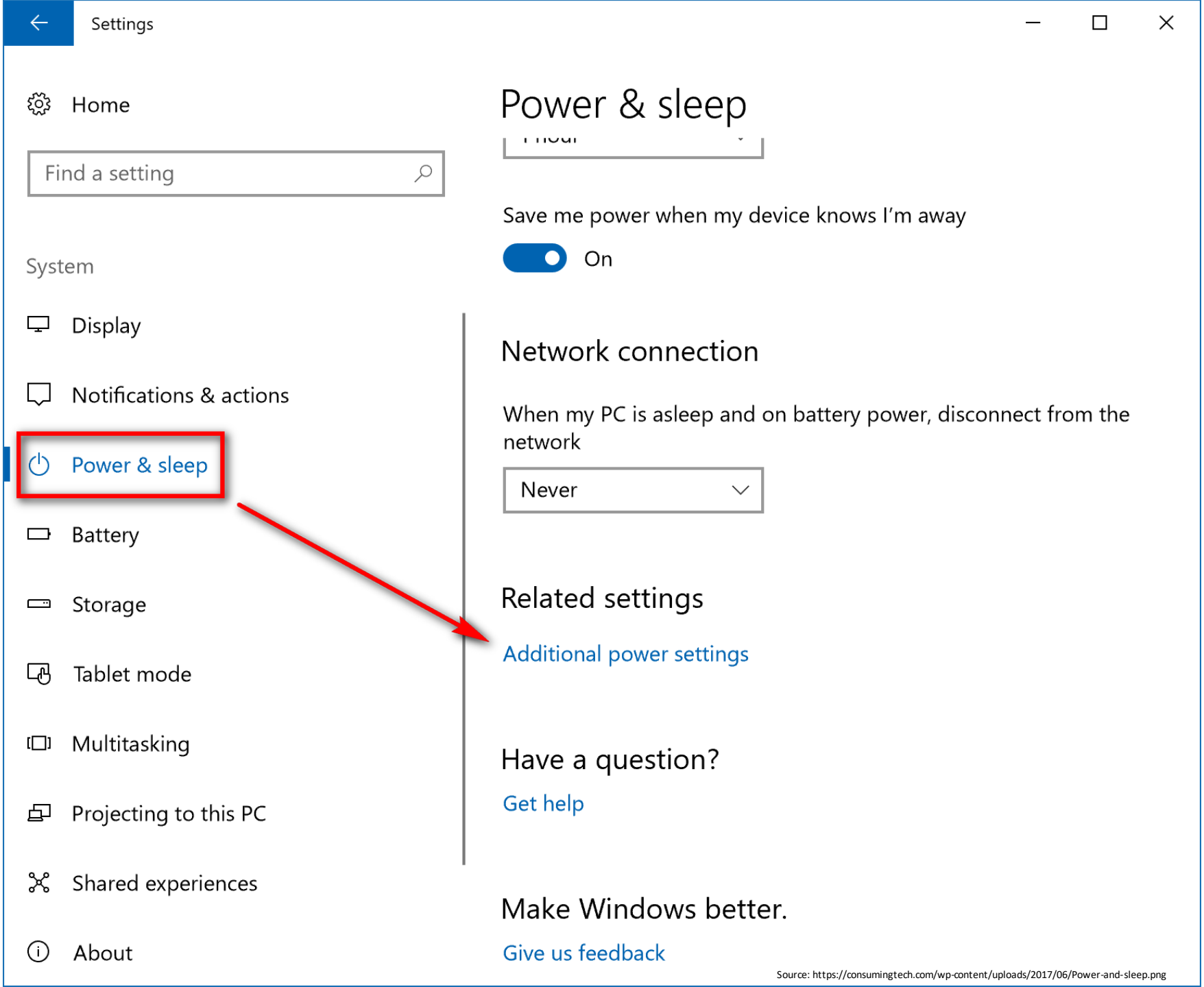
User Domain Knowledge

- Common breakdown: beginner vs. expert
  - Beginners sometimes called "novice users"
  - Experts might be called "power users"
  - Beginners are new to a task, experts have comprehensive knowledge
- In reality it's very much a spectrum
  - Intermediate users are more advanced than beginners but not yet experts
    - Intermediate users sometimes called "average users"



Source: https://business.okstate.edu/site-files/archive/images/support/windows_xp_power_management_2.png

Can we accommodate both types of users?



Source: https://consumingtech.com/wp-content/uploads/2017/06/Power-and-sleep.png

User Roles

- The same system can have groups of users with defined roles
  - e.g. user vs administrator, writer vs. editor
  - Often roles are associated with workplaces

- e.g. Avenue to Learn
  - Students – submit assignments
  - Teaching assistants – mark assignments
  - Instructors – create assignments
  - Administrator – create courses

Tasks

- A **task** is a goal directed behaviour or series of behaviors involving the system
- Tasks are often hierarchal and represented in a tree structure
- Tasks are often made up of subtasks, which themselves can have subtasks
- Sometimes the terms goals, activities and process are used instead of "task"
    - e.g. calling the top-level tasks "goals" for example

Tasks

- **Important:** whenever possible try not to refer to the solution (i.e. the software), especially specifics about the solution, when creating tasks

- Example task: "Finding directions from Hamilton, Ontario to Ottawa, Ontario"

- This is a goal directed activity, but no reference is made to how it is accomplished with the system
  - e.g. "click on the directions button"

Tasks

- Another task: "Click the sort alphabetically arrow"
- Wanting to sort alphabetically might be a subtask of a larger task... but does the UI need to be referenced to express this?
- We're doing task analysis to ***inform*** the requirements, not define them
  - Idea is that we're building a conceptual model of the tasks, more abstract!

Task And Subtask Example

- Search the store for products
  - Filter search results by category
  - Filter search results for sales
  - Sort search results from lowest to highest price
  - Sort search results from highest to lowest price

- In the above, we have a task and 4 subtasks
  - Notice that we don't make reference to specifics of the system and how tasks are achieved

Task And Subtask Example

- Create a new assignment
  - Specify due date
  - Specify group or individual assignment
    - Allow groups to self-select
    - Create groups randomly
  - Select grading rubric
  - Allow multiple submissions or single submissions
- Another example of a task, 4 subtasks, one of which has two subtasks itself
- We can develop a series of task hierarchies like this for all the tasks our system needs to facilitate

Expressing Order Of Subtasks

- We can express the order of subtasks in a simple way with ordered lists

- A cashier will have to enter the items and subtotal them before they can accept payment

- Ringing up a customer
  1. Enter customer items
  2. Total items
  3. Accept item payment

Conducting Task Analysis

- Questions to think about:
  - What makes users begin a task?
    - How do they begin a task?
  - How do users complete a task?
  - What information do users need to complete the task?
  - What tools do users need to complete the task?
  - Do the subtasks have an order?
  - How do users know when a task is complete?

- Try to limit a task's subtasks to 4-8 subtasks
  - If we have more, it suggests we should more layers in our task hierarchy

Task Analysis

- **Hierarchical task analysis** is one form of task analysis
  - In addition to a task hierarchy adds concepts of operations and plans

- **Operations** are the actions performed by the user, they are the subtasks which cannot be broken down further
  - Action does not mean "click X", it's still at a higher level of abstraction, "sort by X", etc.
  - All bottom-level tasks should be operations

- **Plans** are how tasks are to be carried out
  - "First X, then y, then Z"

Hierarchical Task Analysis

1. **DEFINE TASK BEING ANALYZED**, as well as the purpose of the task analysis.
2. **CONDUCT DATA COLLECTION.** Pay particular attention to areas such as technology, machine and team member interaction, decision making, and task constraints to better understand the process.
3. **DETERMINE THE OVERALL GOAL OF THE TASK.** This should be in place at the top of the task hierarchy.
4. **DETERMINE TASK SUB-GOALS.** Decompose the overall goal into corresponding subgoals. Together, these subgoals should comprise the tasks necessary to accomplish the overall goal.
5. **PERFORM SUB-GOAL DECOMPOSITION.** Subgoals should be further broken down into additional subgoals and operations. Continue this process until you reach an appropriate operation, which specifies the action that actually needs to be done to accomplish the goal. There should always be an operation at the bottom level of any branch within an HTA.
6. **DEVELOP PLANS ANALYSIS.** After describing all subgoals and operations, add the plans. Plans explain how a goal should be accomplished. These may be in the format "do A, then, do B, then do C", or "do A, or do B and do C"

Source: https://digital.ahrq.gov

Hierarchical task analysis

- Checkout
  - Enter address
  - Select shipping information
  - Enter promotion code
  - Provide payment information
    - Select Visa or Mastercard
    - Enter expiry date

- **Operations:** all tasks except "Checkout" and "Provide payment information" would be operations

- **Plan:** "Enter address, followed by select shipping information, followed by …"

Now what data collection practices can we use to help inform and perform all this analysis?

Background Research

- Basic information collection and research: existing system documentation, organizational documents (e.g. how a process is done currently), books, articles, web resources, academic journal articles, videos, social media, "Google-fu", etc.
- Try to understand the domain: related systems, experts, how similar problems have been solved
- Try to understand the system as is: feature documentation, change requests, workflows
- Possible creation of very preliminary requirements
- Allows for suggestion of potential solutions to stakeholders

Surveys

- Low cost/time commitment per user allows scalability to a large number of users
- One free web survey can scale to tens of thousands of users...
  - Google Form
  - Survey Monkey
- Distribution via e-mail or other online means can be instant

Surveys

- **Important:** surveys can be easily made anonymous, allowing for (much) more honest feedback
  - Ever heard the expression "nobody wants to call the baby ugly"?
  - Think about employee stakeholders that don't have the influence or importance of employer stakeholders
- Quantitative with objective data points, average scores, etc.
- Questions can be reviewed for bias and other issues, more difficult to do this with interviewers
- Survey results can guide interviews done later, probing deeper with follow-up questions into points of concern

Interviews

- High cost/time commitment per user
- Far less anonymity, if at all
- But can be exploratory, possibility of follow-up questions
- Qualitative... can see the emotion in a user's face!
  - Good for learning about user feelings, attitudes
- Informally done interviews tend to be one of the most common ways to gather requirements in practice!

Interview Basics

- Prepare questions in advance
- Meet one on one and in groups
- Discuss answers
- Document all answers
- Follow up if needed

Interview Question Types

- **Open-ended questions** are questions which encourage discussion and explanation
  - e.g. "How do you do this right now?"

- **Closed questions** are questions that look for specific data
  - e.g. "How many forms do you process in a day?"

Interview Types

- A **structured interview** is focused on getting answers to known questions, focused discussion on pre-set topics.

- An **unstructured interview** allows for free discussion on the system, the perceived issues and potential gains. May explore previously unknown topics.

- Note that effective interviews will most often combine both types of questions and both types of structure.

Interview Preparation

- Organize any useful artifacts
  - Current system documentation
  - Survey results
  - Scenarios
  - Prototypes
- Select stakeholders by type of info to gather
  - Determine cross-section of stakeholders to give a full overview
  - Likely not feasible to interview all stakeholders
- Determine individual vs. group interviews
  - Group interviews can save time
  - Individual interviews allow more in-depth questioning

Interview Preparation

- How to determine appropriate cross-section of interviewees?
  - Need operational stakeholders (i.e. users) who will be using system in multiple ways
  - e.g. Operators, admins, guests
- Document a plan for the interview, like an experimental procedure
  - What questions to ask, when, free-form discussion, etc.
  - How will data be recorded? Audio? Transcription? Memorization?
  - How formal this is can scale from (possibly mental) bullet points all the way to a formal multipage document

Basic Interview Plan Example

- **Restaurant owner/manager:** Discuss project constraints, budget, timeline, available tools. Ask if there are any extra features needed, such as ability to support new franchise locations.

- **Head chef:** Discuss physical constraints of the kitchen, how to determine if supplies are sufficient and how to estimate time to cook.

- **Line cook:** Discuss number of dishes prepared in a night, what issues can come up in a normal night, what are the busiest times.

- **Store manager:** Discuss number of employees, any plans to expand or add new dishes. What are the open hours.

- **Regular customer**: How do you normally place your orders, how can it be improved

Focus Groups

- A **focus group** is a type of group interview with a group of participants with some trait or experience in common, *where interaction between participants is encouraged*

- The more sensitive the nature of the discussion, the more it is recommended that focus group members share characteristics
  - Improves honesty of the feedback
  - e.g. would you be more honest around fellow employees or your manager?

Focus Groups

- The biggest value is the group discussion that can occur, probing deeper into "why" questions
  - A moderator is typically key to facilitating this discussion
- Different formats can be used, but generally a relaxed setting with open questions are used
- Potential problem: groupthink
  - People will "go with the group" rather than state their own opinions
  - The effects of this cannot be understated, even in small groups with people that don't know each other

Ethnographic Research

- **Ethnographic research** is a qualitative user research method based on observing users in their real-life environment

- e.g. embedding yourself on the factory floor and observing all the different roles, what they involve

- Ethnographic research can be especially effective for discovering user roles, workflows and tasks

Ethnographic Research

- Ethnographic research can be **passive** where observations are made but no interaction occurs
  - Passive observation may be better at evaluating how current systems work (or even a prototype system)
- Ethnographic research can also be **active** where questions are posed to those being observed
  - Active observation can be more effective at understanding processes
- Methods can be combined, and ethnographic research is a field onto itself!

Personas

- Personas
  - Creation of a representative user(s) based on available data and user interviews
  - Personal details of a persona are fictional, but data about users that is used to create it is not!
- How can personas help?
  - Aggregate data turned into common reference point
  - Thinking through: how would "Jane" user this system?
  - Argument: personas help us to empathize
- Empirical evidence for personas as being effective
  - Studies have come under criticisms though
  - Some say personas unscientific because not real person

Emilie Anderson

Xtensio



"The best preparation for tomorrow is doing your best today."

Age: 27
Work: Journalist
Family: Husband & Daughter.
Location: Newcastle, NSW, AU.
Character: Caring, active & passionate.

Personality

Introvert	Extrovert
Thinking	Feeling
Sensing	Intuition
Judging	Perceiving

Smart | Ambitious | Active | Good

Frustrations

- Not being able to eat yummy food without being sick.
- Exhaustion.
- Not being able to go out with friends for dinner.

Goals

- Live a happy healthy life.
- Learn more about her condition to understand.
- Lose 5kg.

Bio

Emilie is a young woman who suffers from Irritable Bowel Syndrome. She has a young family and needs all the energy she can get. Emilie is a vibrant and loving woman she loves to run and eat healthy but is so limited on what she can eat.

Motivation

- Incentive
- Fear
- Growth
- Power
- Social

Brands & Influencers

LUSH FRESH HANDMADE COSMETICS · running bare · HARRIS FARM MARKETS · MECCA MAXIMA

Preferred Channels

- Traditional Ads
- Online & Social Media
- Referral
- Guerrilla Efforts & PR

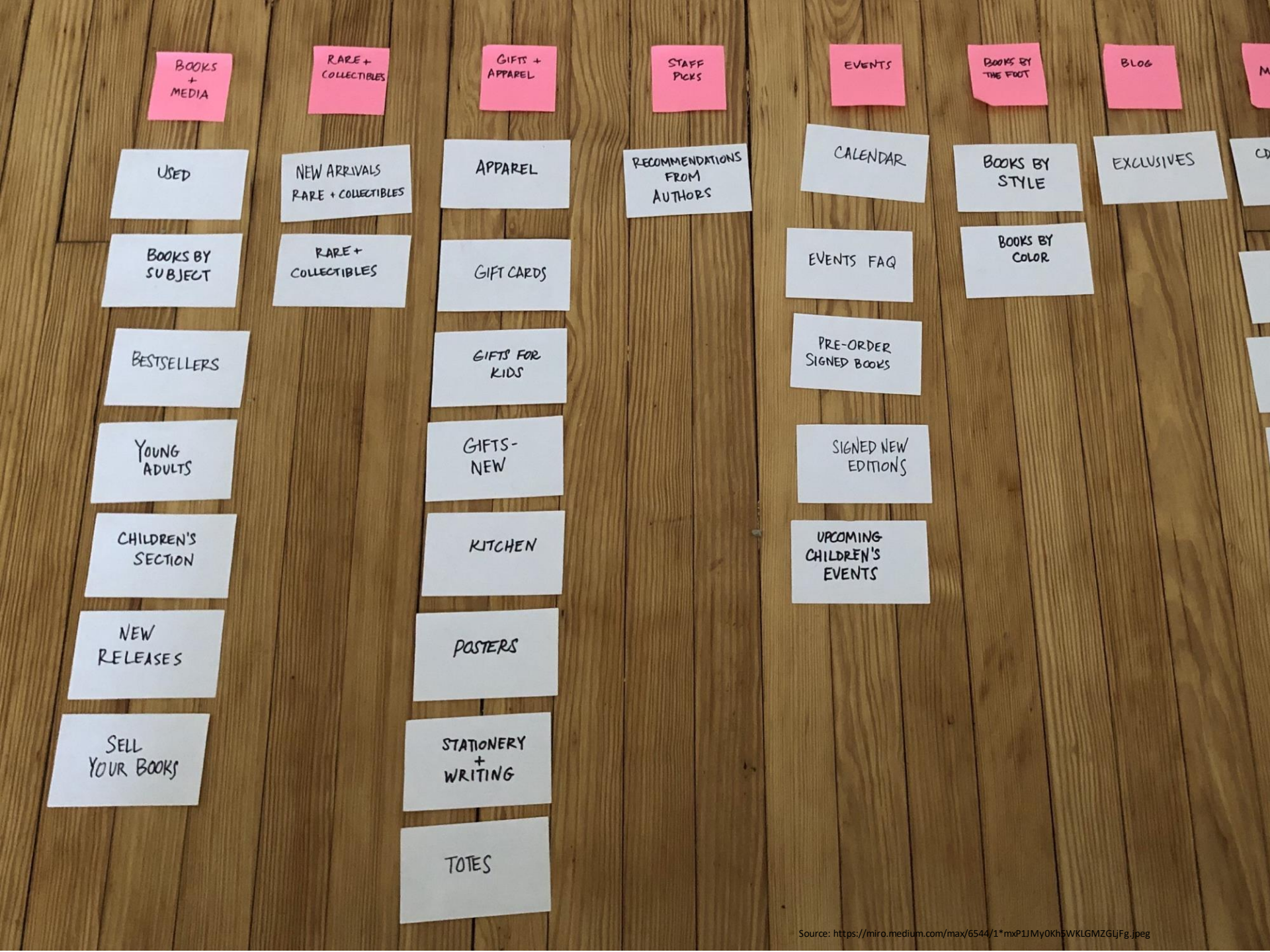
Source: https://problem5olver.files.wordpress.com/2018/09/user-persona-e1535863445852.jpg?w=1400&h=9999

Card sorting

- Card sorting helps us to answer the question:
  - What information is or ***should*** be associated with what other information

- Helps us to arrange website menus, workflows, navigations
  - Information architecture: https://en.wikipedia.org/wiki/Information_architecture

- Cards are informally arranged into a tree structure to represent data and relationships



Source: https://miro.medium.com/max/6544/1*mxP1JMy0Kh5WKLGMZGLjFg.jpeg

Source: https://i.pinimg.com/originals/be/33/17/be331764d91c99a7184e9e0427b2de34.jpg



Card Sorting Process

- Participants in a card sorting process are typically users
- A **closed card sorting** process asks participants to sort cards under predefined headings
- An **open card sorting process** asks participants to group cards into categories that make sense and to assign these categories names
- Card sorting takes many forms, e.g. done individually or in groups
  - With groups, group-think can play a role (false consensus)

Current System Data

- If the application we are building is replacing a current software system or non-software process, any data related to that system may be helpful

- For example...
  - Log data of current usage
  - Remote monitoring of current usage (e.g. Zoom with a shared screen)

- This data can also be analyzed to inform our requirements

Brainstorming

- **Brainstorming** is a group creativity technique that involves creating a list or lists of ideas spontaneously contributed by members

- For example, gathering stakeholders in a room to brainstorm a list of required options for a feature

- Goal is quantity over quality, quality can come later

- Many styles of conducting brainstorming, e.g. giving all participants a sheet to fill out then share results v.s. gathering responses on a whiteboard

Other Tools

- Other practices such as scenarios and prototypes overlap between data collection and representing requirements

- A **scenario** represents the interaction of a user or different types of users with a system

- A **UI prototype** is a representation of a user interface for a system

Other Tools

- We use these tools to help define requirements
- But we also develop them working with users, e.g. possibly together in the same room
- We also use them as a discussion point with users during interviews and surveys
  - To verify a shared understanding of user needs
  - To analyze possible alternatives

Other Tools

- We'll over these tools separately since they are both big topics
- There are multiple types of scenarios
- There are multiple types of UI prototypes
- Despite overlaps, we are now moving towards specifying the requirements, not just gathering